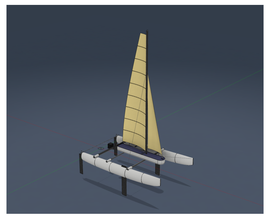
Designing a 1 m RC Foiling Beach Catamaran: CAD, Velocity Prediction, and a Build Plan

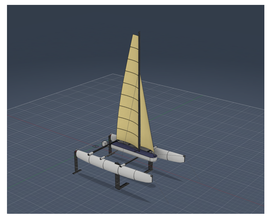
Filip Maleev
Colorado College, Mathematics | September 2026

The question

Can a 3D-printed 1 m radio-controlled catamaran look and sail like a real beach cat (Hobie 16, Nacra 17), **fly on hydrofoils**, and still be **controllable**? The answers had to be known before building.



Build-first version (daggerboards)



Foiling version (v2.2 L-foils, T-foil rudders)

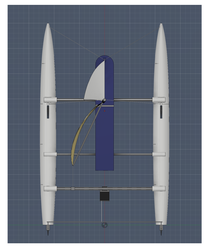
Design (Autodesk Fusion, scripted through its API)

- **Hulls:** 8 elliptical stations, 1000 mm, printed as 0.8 mm PETG shells. Bulkheads bonded in at every print joint make each segment **a sealed compartment**.
- **Rig from real cats:** square-top main with 8 full-length battens, rotating wing mast, forestay bridle, traveler and jib car.
- **Scale effect:** righting moment $\propto \lambda^4$ but heeling moment $\propto \lambda^3$, so sail area was sized by analysis, not by geometric scaling.

	Hobie 16	Nacra 17	**This boat**
Mast / LOA	1.60	1.71	**1.37**
Main : jib	2.7	3.7	**3.2**
Sail area / LOA^2	0.74	0.68	**0.40**

Sail trim on sliders

A custom Fusion add-in maps the three real-cat controls to three sliders and three RC servos: **jib sheet**, **main traveler**, and **mainsheet tension** (twist 30° → 3°, depth 13 % → 8 %). The sails are parametric lofts that re-solve as you slide.



Right: reaching trim seen from above, boom eased 25°, the top of the main twisted off.

Method: a purpose-built velocity prediction program

CAD geometry + mass properties (Fusion API)
↓
Hydrostatics: station integration (0.20 % of CAD volume)
↓
Resistance: ITTC-57 friction + **Michell wave integral** + appendages
↓
Foil lift: thin-airfoil α_{L0} + lifting line
↓
Sails: C_L, C_D model, flattened to stay under 90 % of righting moment
↓
VPP: speed where drive = drag; controllability checks

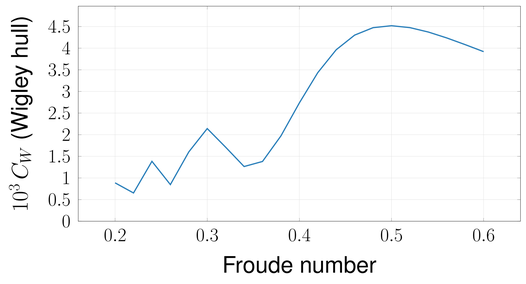
Michell's thin-ship integral, with a catamaran interference factor F_c:

$$R_W = \frac{4\rho g^2}{\pi U^2}\int_1^\infty \left(I^2 + J^2\right)\frac{\lambda^2 F_c(\lambda)}{\sqrt{\lambda^2-1}}\,d\lambda$$

$$F_c = 4\cos^2\!\left(\tfrac{g}{U^2}\lambda\sqrt{\lambda^2-1}\,\tfrac{s}{2}\right)$$

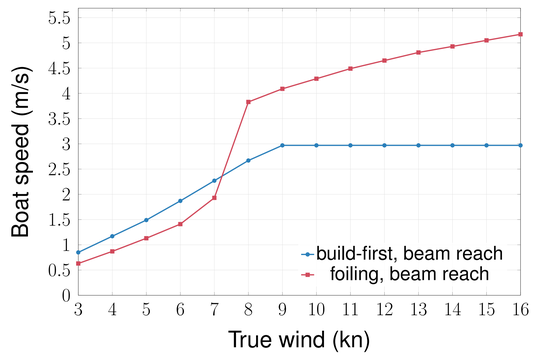
Integrated exactly in x and z on each panel (Filon-type), so it stays accurate at low Froude numbers.

Validation



Grid refinement changes the result by less than **0.03 %**, and the hump at $Fn \approx 0.3$ and the main hump at $Fn \approx 0.5$ both appear. A hull-drag regression test guards every code change.

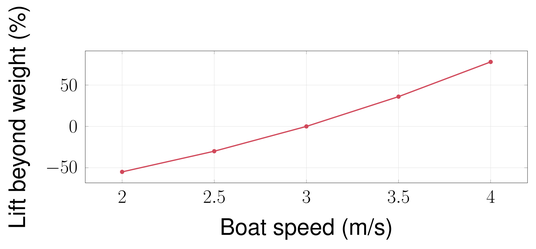
Results: speed and take-off



- Build-first: **2.7 m/s** in 8 kn of wind on a beam reach.
- Foiling: takes off at **2.0 m/s** from about **7.1 kn**, below the 8–9 kn where the rig must be depowered.
- Sensitivity (sail C_L ±15 %, foil lift −20 %, mass ±10 %, sail-drive couple ×0.5–1.5): take-off between **6.75 and 8.75 kn**.

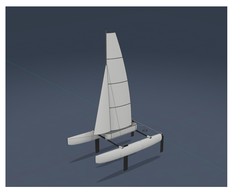
Results: can it be controlled?

- **Rudders (build-first):** need at most **2.6°** of ±18° throw, with ≥78 % margin to stall, even with the windward rudder out of the water.
- **Foil pitch:** stable, with the CG 32–84 mm ahead of the neutral point.
- **Bows (pitchpole):** v1.3 hulls add forward volume: +23 % bow-down margin for +0.4 % drag. The VPP now eases the sails for pitch as well as heel.
- **Foil heave: NOT stable.** Fixed-angle foils lift **+78 %** more than the boat's weight at 4 m/s, so the boat would climb, breach and crash.



v2 fix, sized: a bow wand on each hull drives a 25 %-chord flap on that hull's foil. Heave becomes stable and over-damped ($\zeta = 1.4$, 2.1–5.3 Hz) using at most 13.0° of flap across the flying range. **Build order:** mini, then the 1 m boat without foils, then foils with wands.

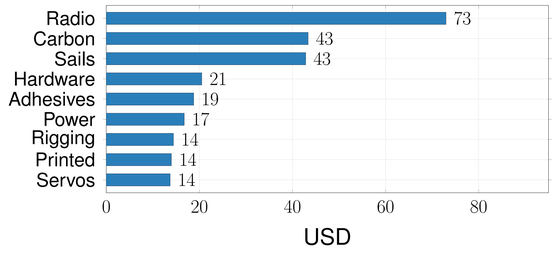
First build: RC Mini Cat 600



Same lines, rig and printed construction at 0.6 length, but hull sections scaled 0.68, because the electronics do not shrink. One winch with a **stepped drum** (30 / 16.2 mm) pulls both sheets, sized so both reach full ease in the same 1.63 turns.

- 755 g sailing weight; one hull alone floats 2.05× its weight
- 2.1 m/s predicted in 8 kn; at most 1.6° of rudder needed

What it costs (USD, prices read 23-24 Sep 2026)



$257.37 in parts for the mini | **$381.02** for the 1 m boat (+$99.45 foils)

Every price was read on a retailer page. The build manual has the full list, 1:1 sail patterns and 10 build steps.

Checked in CAD, next on the water

- Servo rails redrawn from the makers' drawings.
- A 9-trim clearance sweep moved the pod forward onto a light third beam.
- Flight simulator with pitch trim, 306,180 foil designs: positive-incidence elevators would pitch the boat nose-down and out of the water. **v2.2** (in CAD): elevators at −1°, 13 mm lower; flies level at 12 kn.
- **Next:** build the mini, then the 1 m boat; tow-test the v2.2 foils, then fly.

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